

Table 1. Summary MGSM multilingual-transfer results for Qwen3-4B and Gemma-3-4B-Instruct, macro-averaged equally across five languages within each model. Higher is better for every column.

Model	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Original	41.6	0.00	1.76	1.47
	S-PID	0.0	<u>1.98</u>	0.00	0.00
	A-LQR	22.4	1.81	1.51	1.08
	H_∞ (ours)	32.2	1.44	1.66	1.25
Gemma-3-4B-Instruct	S-PID	<u>0.0</u>	0.00	<u>0.00</u>	<u>0.00</u>
	A-LQR	<u>0.0</u>	0.21	<u>0.00</u>	<u>0.00</u>
	H_∞ (ours)	0.0	0.29	0.00	0.00